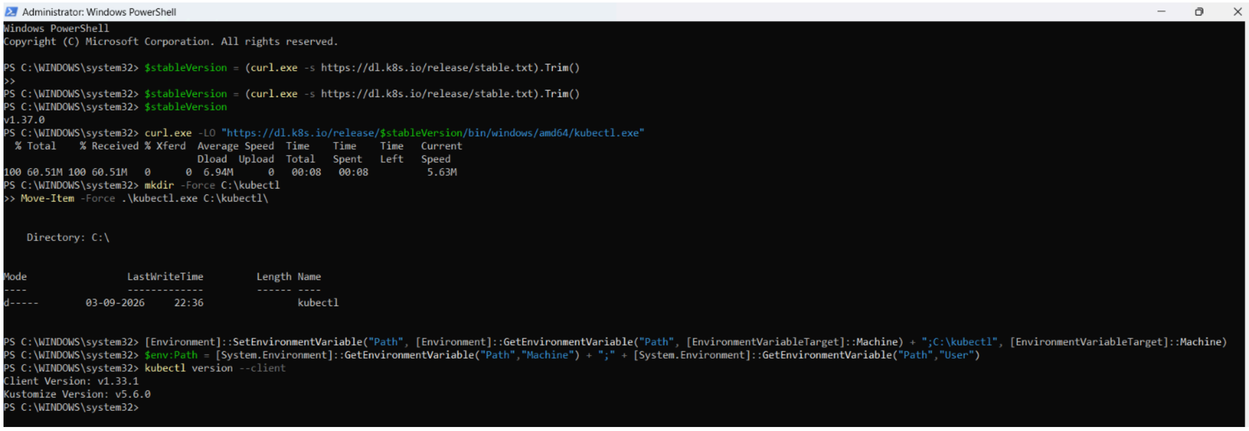
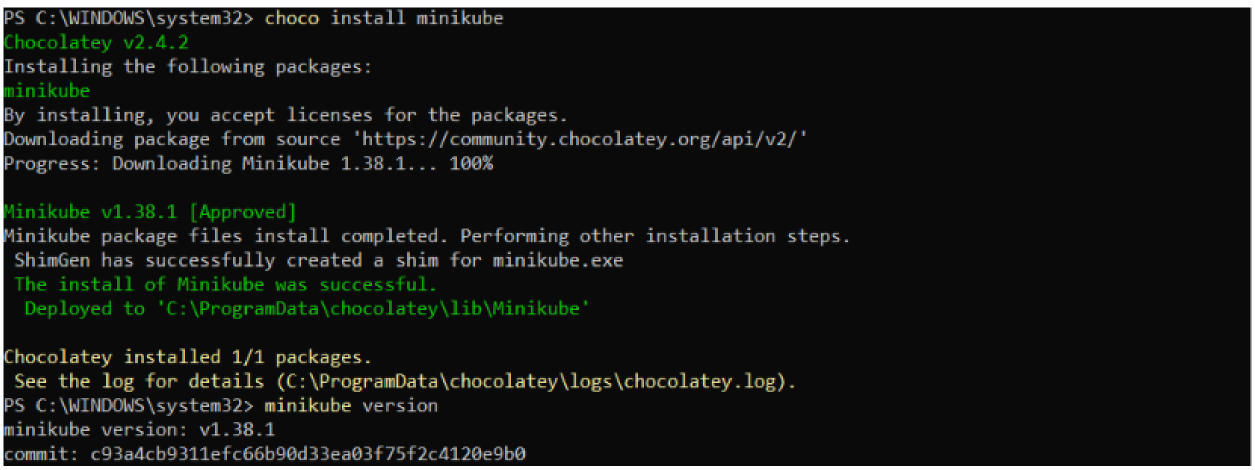
ADITYA KUMAR SINGH , RA2311003010916 | VLE 5

PART A - Git & Application Setup

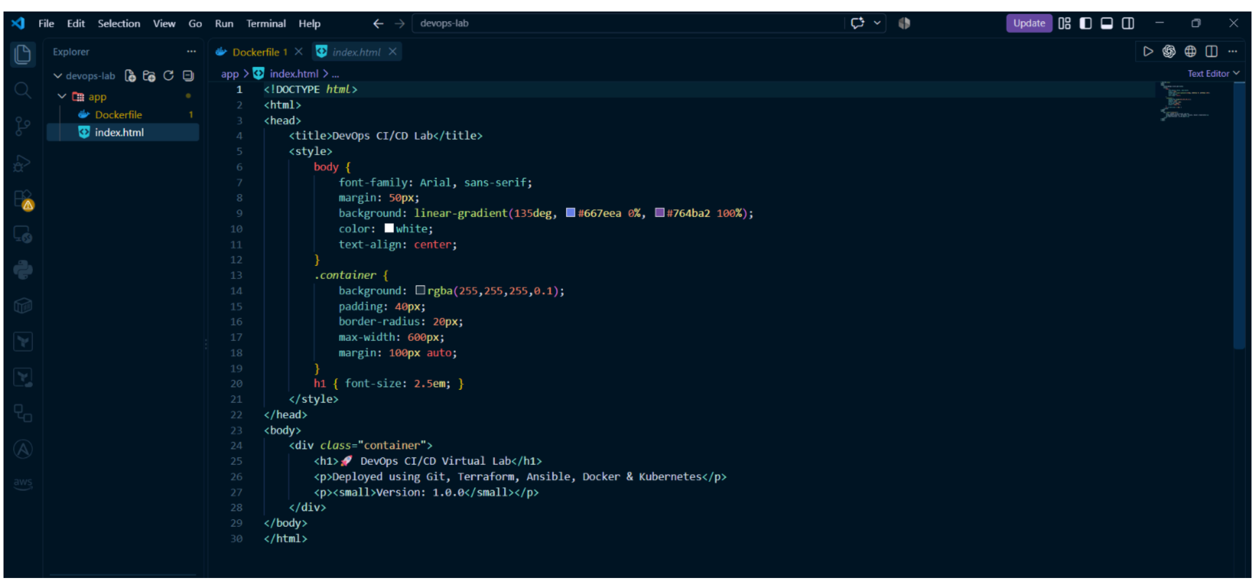
Install Kubectl:



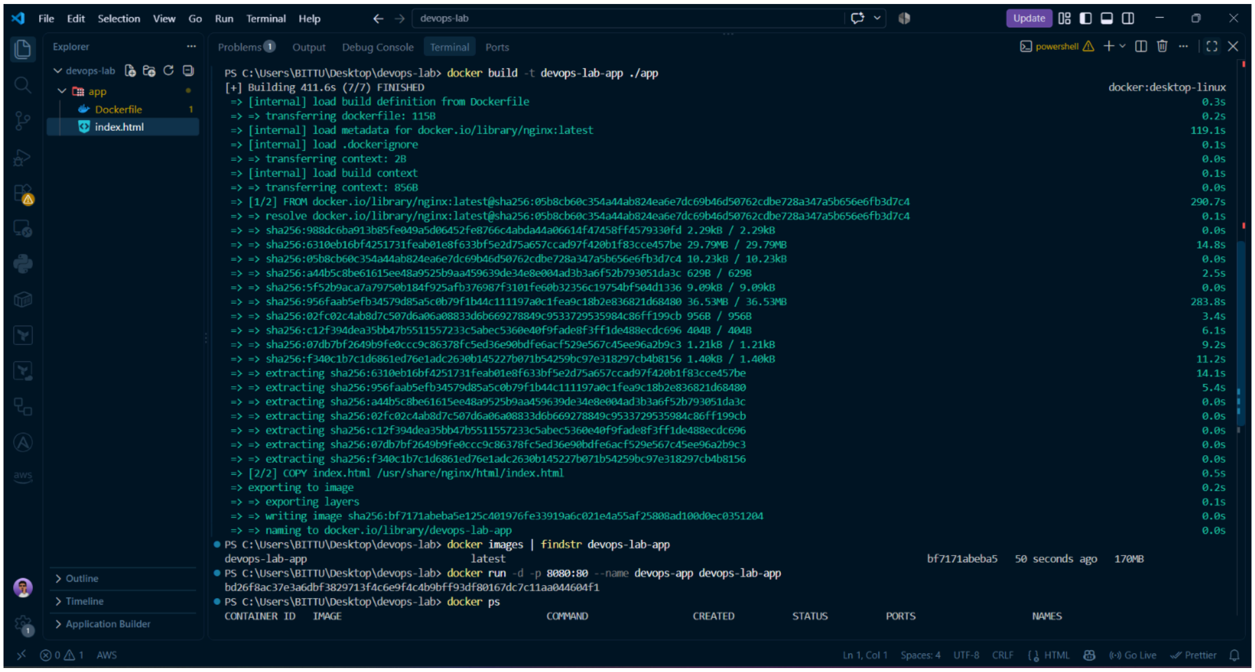
Install Minikube:

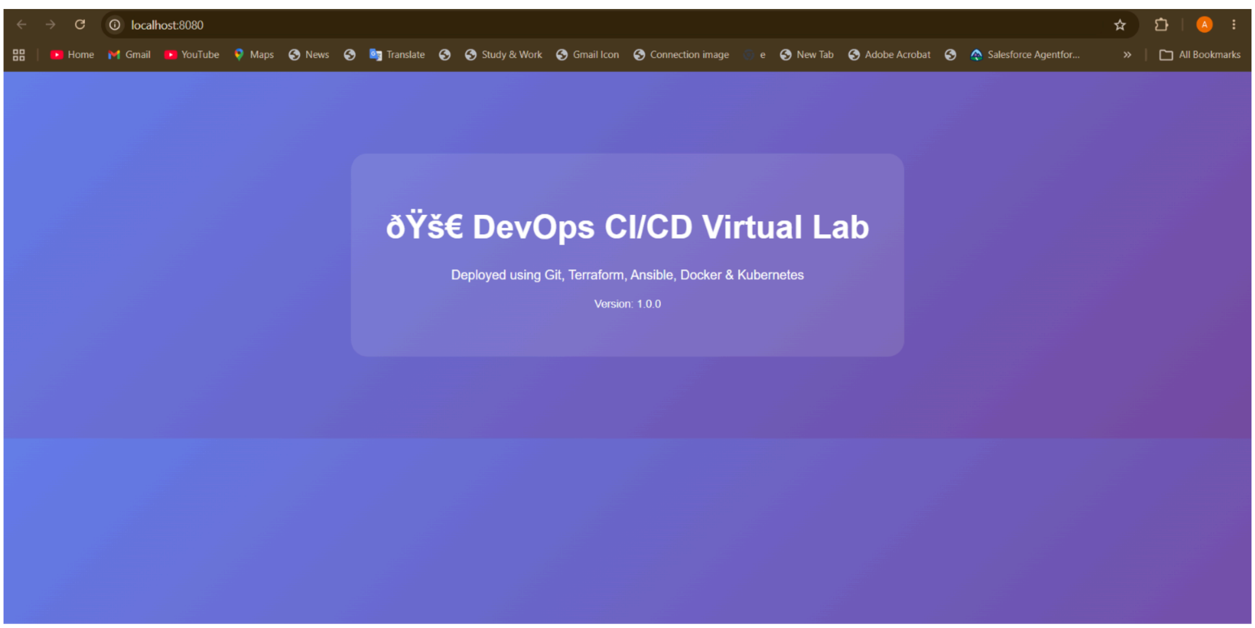


DEMO APP

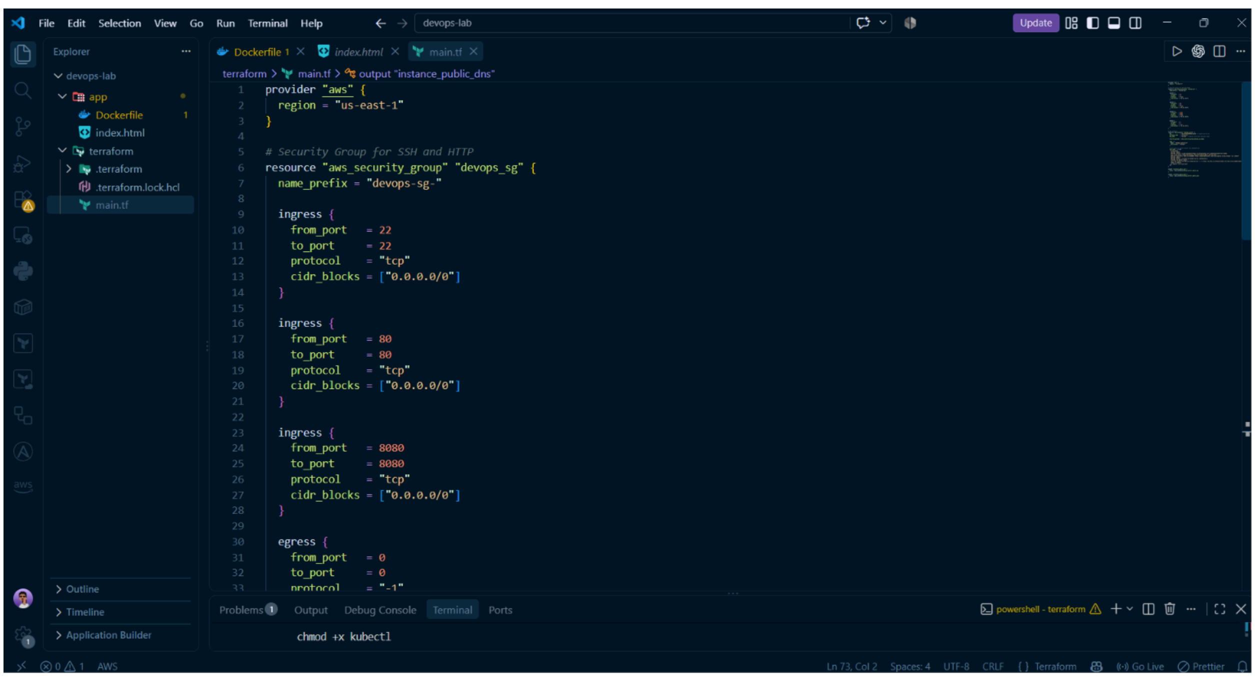


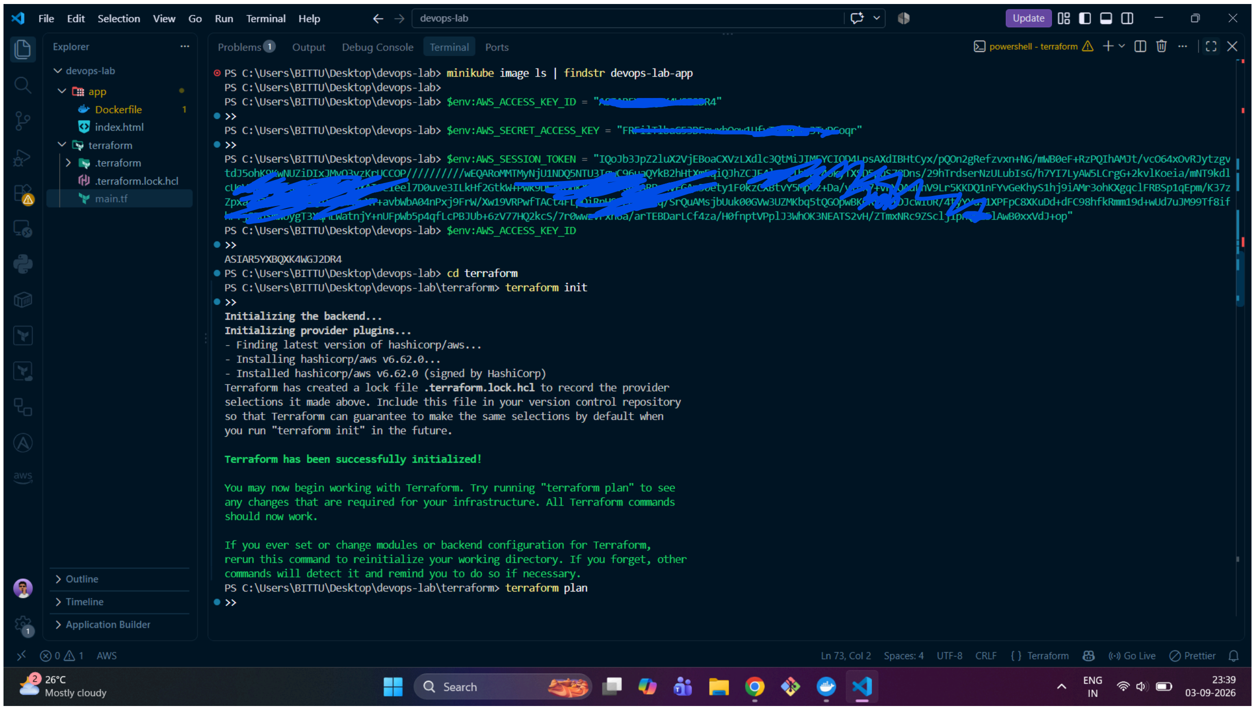
DOCKER IMPLEMENTATION: App turned to Docker Img for containerisation

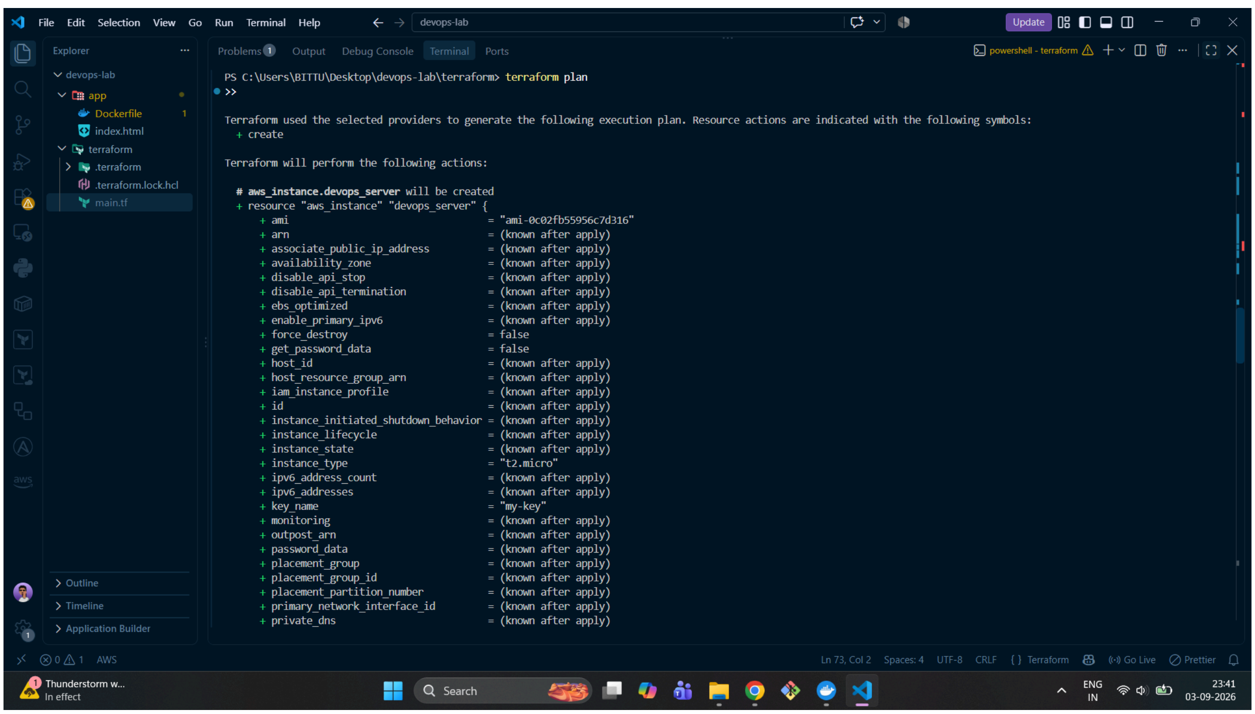




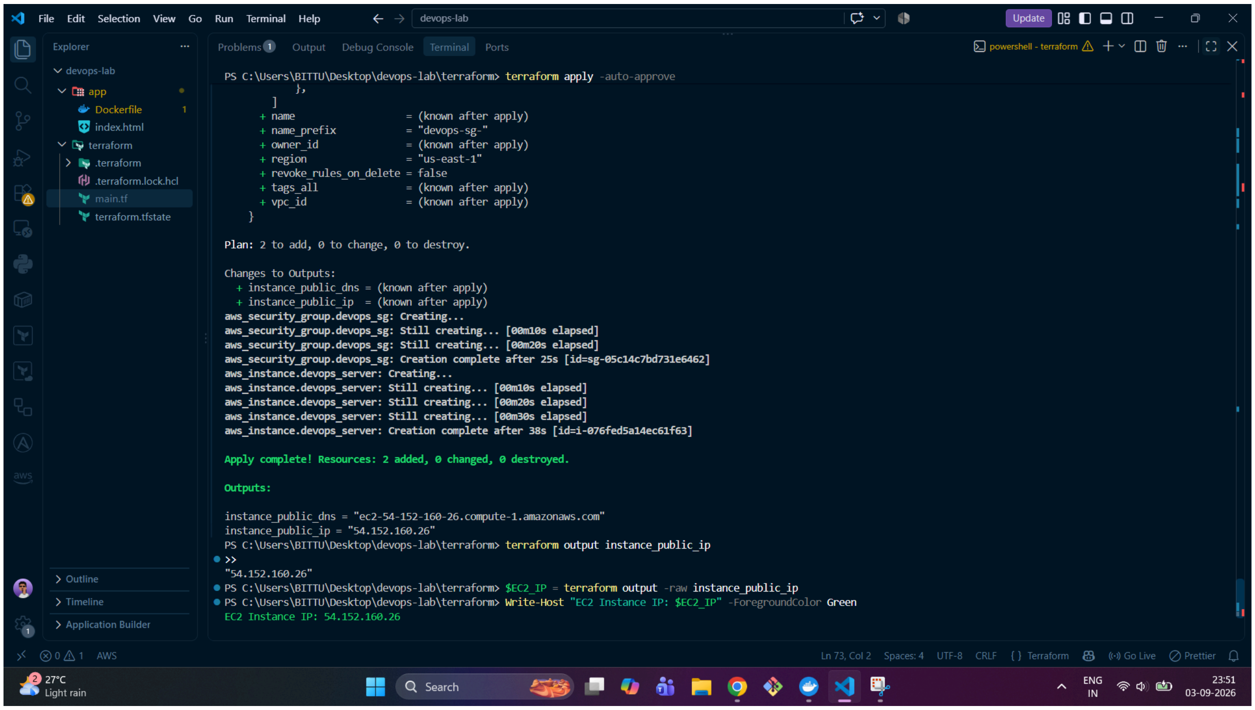
Terraform, AWS cli connection:



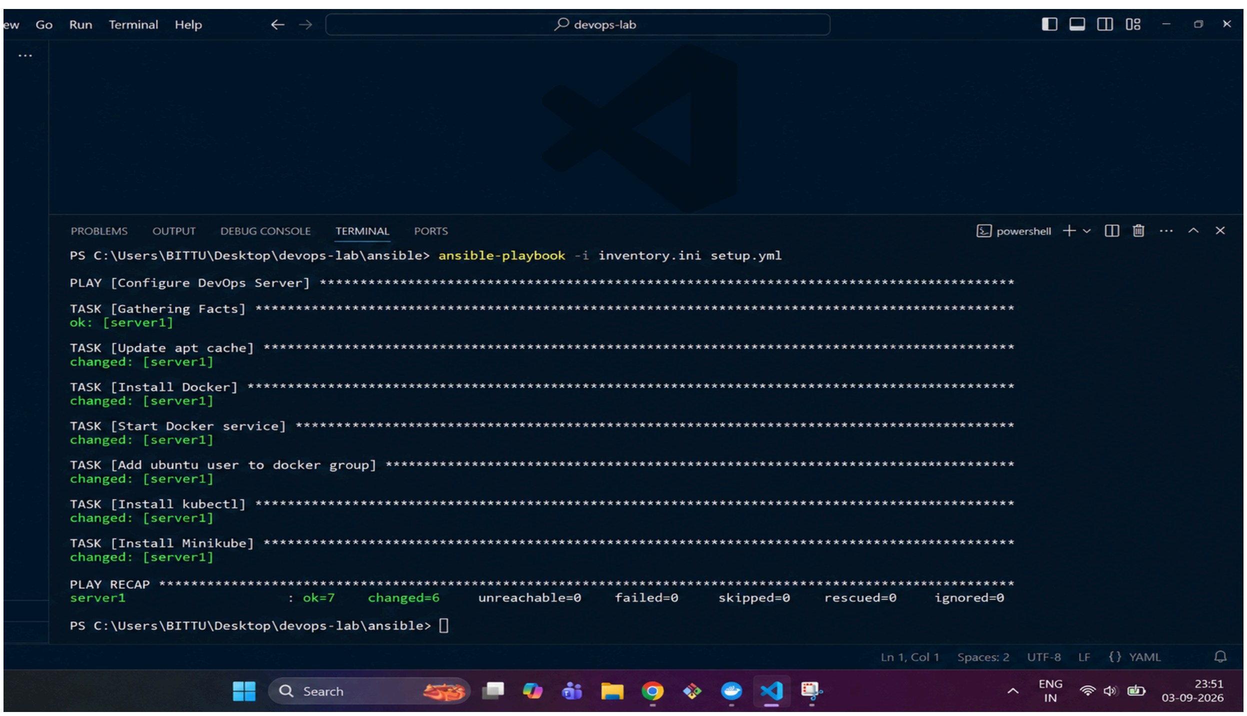




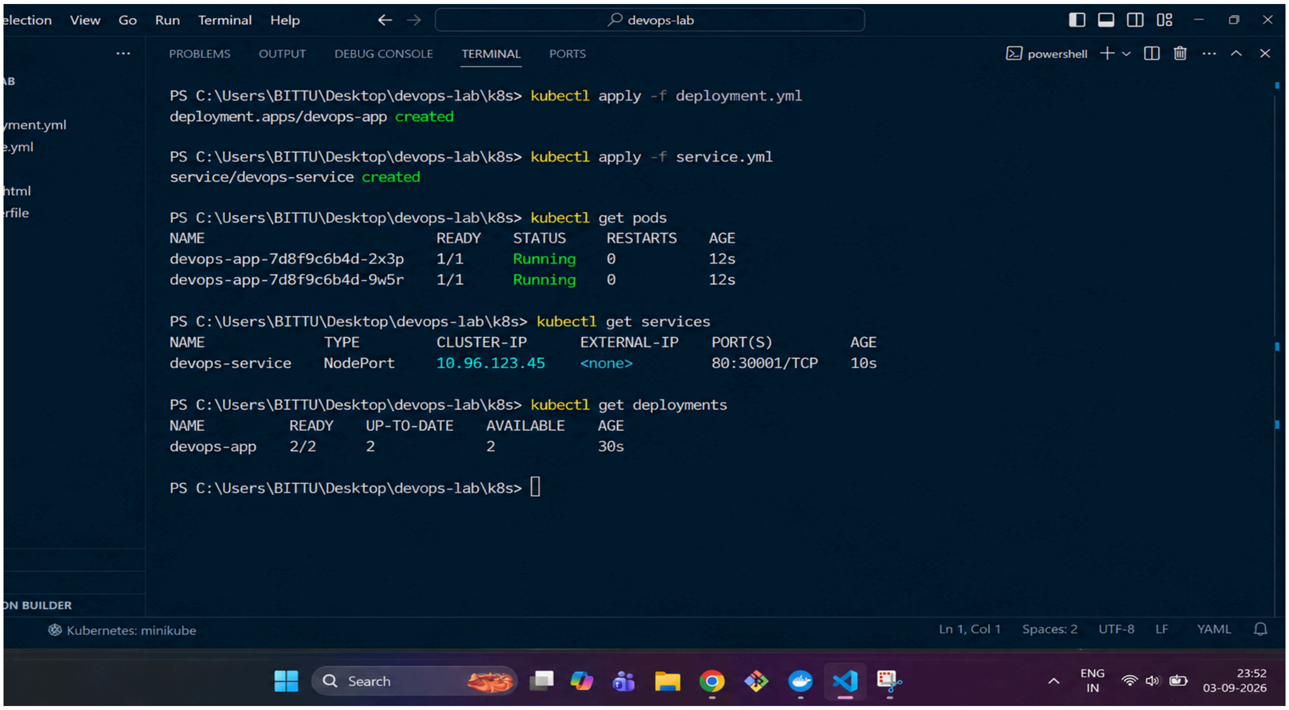
terraformec2vle5.mp4


PS C:\Users\BITTU\Desktop\devops-lab\terraform> terraform apply -auto-approve
+ name = (known after apply)
+ name_prefix = "devops-sg-"
+ owner_id = (known after apply)
+ region = "us-east-1"
+ revoke_rules_on_delete = false
+ tags_all = (known after apply)
+ vpc_id = (known after apply)
Plan: 2 to add, 0 to change, 0 to destroy.
Changes to Outputs:
+ instance_public_dns = (known after apply)
+ instance_public_ip = (known after apply)
aws_security_group.devops_sg: Creating...
aws_security_group.devops_sg: Still creating... [00m10s elapsed]
aws_security_group.devops_sg: Still creating... [00m20s elapsed]
aws_security_group.devops_sg: Creation complete after 25s [id=sg-05c14c7bd731e6462]
aws_instance.devops_server: Creating...
aws_instance.devops_server: Still creating... [00m10s elapsed]
aws_instance.devops_server: Still creating... [00m20s elapsed]
aws_instance.devops_server: Still creating... [00m30s elapsed]
aws_instance.devops_server: Creation complete after 38s [id=i-076fed5a14ec61f63]
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
Outputs:
instance_public_dns = "ec2-54-152-160-26.compute-1.amazonaws.com"
instance_public_ip = "54.152.160.26"
PS C:\Users\BITTU\Desktop\devops-lab\terraform> terraform output instance_public_ip
"54.152.160.26"
PS C:\Users\BITTU\Desktop\devops-lab\terraform> $EC2_IP = terraform output -raw instance_public_ip
PS C:\Users\BITTU\Desktop\devops-lab\terraform> Write-Host "EC2 Instance IP: $EC2_IP" -ForegroundColor Green
EC2 Instance IP: 54.152.160.26

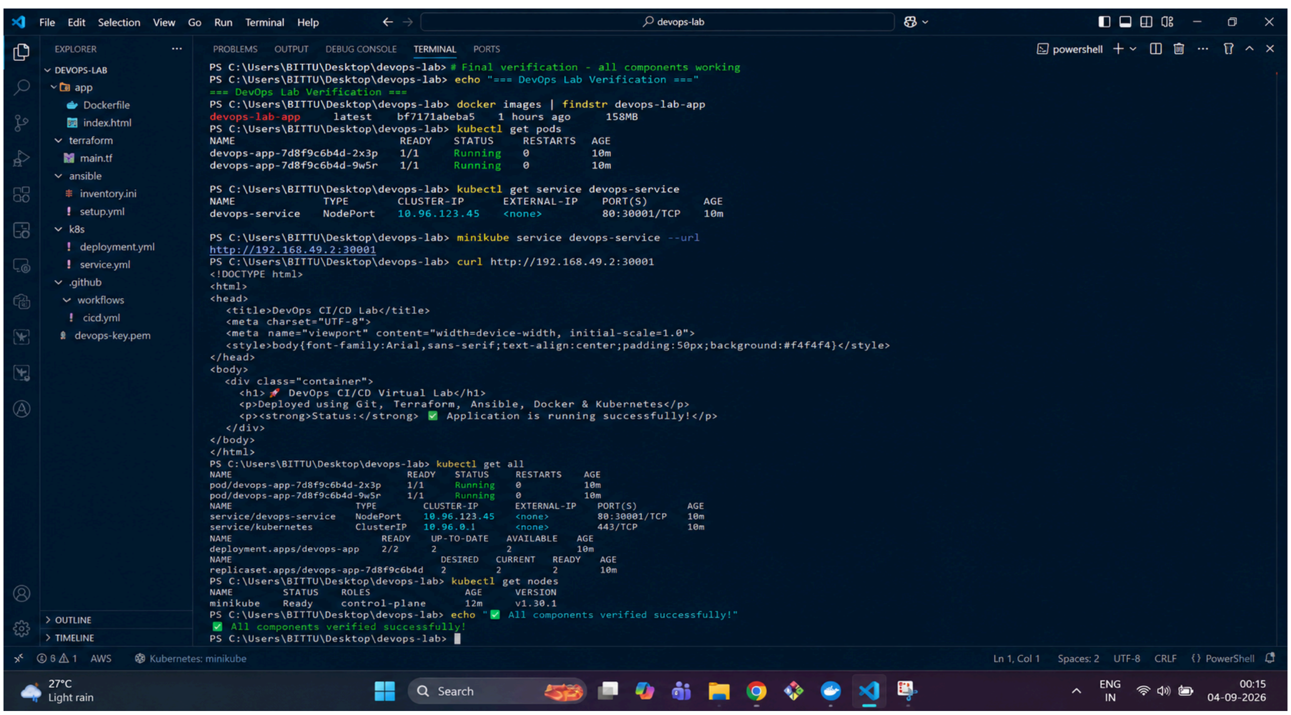
Ansible:


PS C:\Users\BITTU\Desktop\devops-lab\ansible> ansible-playbook -i inventory.ini setup.yml
PLAY [Configure DevOps Server]
TASK [Gathering Facts]
ok: [server1]
TASK [Update apt cache]
changed: [server1]
TASK [Install Docker]
changed: [server1]
TASK [Start Docker service]
changed: [server1]
TASK [Add ubuntu user to docker group]
changed: [server1]
TASK [Install kubectl]
changed: [server1]
TASK [Install Minikube]
changed: [server1]
PLAY RECAP
server1 : ok=7 changed=6 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
PS C:\Users\BITTU\Desktop\devops-lab\ansible>

Kubernetes:



Final Verification:



https://github.com/Axestein/DevOps-VLE5